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**From Discriminative Baselines to Large Multimodal
Models: A Comparative Study of CNN-BERT,
BLIP-2, and Qwen3-VL for Medical Visual Question
Answering**

FINAL REPORT

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ABSTRACT

Medical Visual Question Answering (Med-VQA) aims to assist clinicians by interpreting complex radiological images through natural language interactions. However, traditional approaches have largely treated this as a classification task restricted to a fixed vocabulary, failing to capture the nuance required for real-world diagnostics. This study systematically evaluates the technological evolution of Med-VQA by going through three distinct architectural eras: a discriminative baseline (ResNet-50 fused with BERT), a generative Vision-Language Model (BLIP-2), and a state-of-the-art Large Vision-Language Model (Qwen-VL) equipped with dynamic resolution. Using the VQA-RAD dataset, we identified a critical resolution trap where the generative BLIP-2 model (57.76%) surprisingly underperformed the simpler discriminative baseline (59.48%) on closed-ended tasks due to fixed 224x224 image compression. In contrast, Qwen-VL, utilizing dynamic resolution to process images at native aspect ratios, achieved a dominant 77.59% accuracy.

Beyond accuracy, our analysis of training dynamics reveals a massive confidence gap, with Qwen-VL achieving an order-of-magnitude lower validation loss (0.64 vs. 3.78) compared to the baseline. Crucially, our analysis of clinical safety reveals that Qwen-VL reduced the rate of false negatives by 50% compared to BLIP-2 (13 vs. 26 cases) and significantly mitigated object-level hallucinations (e.g., organ misidentification). However, we identified a remaining challenge of plausible hallucinations, where the model generates medically coherent but incorrect differential diagnoses. While Qwen-VL demonstrates superior semantic understanding with a ROUGE-L score of 43.79, we observe an unavoidable accuracy-latency trade-off, with inference speeds approximately 28% slower than lower-resolution models. These findings confirm that breaking the visual resolution bottleneck is essential for the next generation of reliable radiological AI assistants.

Keywords: *Medical VQA, Large Vision-Language Models, Dynamic Resolution, Qwen-VL, BLIP-2, Deep Learning*

1. INTRODUCTION

1.1 Background

The modern healthcare landscape creates a large amount of medical imaging data, including X-rays, CT scans, and MRIs. While these images contain critical diagnostic information, the manual interpretation process is time-consuming and prone to human error due to radiologist fatigue. Medical Visual Question Answering (Med-VQA) has emerged as a promising solution to this challenge, functioning as an automated second opinion. Unlike traditional Computer-Aided Diagnosis (CAD) systems that output a single label (e.g., "Pneumonia: Positive"), Med-VQA systems combine Computer Vision (CV) and Natural Language Processing (NLP) to understand specific clinical questions and provide contextualized answers that simulate a natural dialogue between a physician and an AI expert.

1.2 Problem Statement

Despite the promise of Med-VQA, current methodologies face three primary limitations that hinder clinical adoption:

1. **The Paradigm Conflict:** Early Med-VQA systems utilized discriminative architectures that treat answer generation as a multi-class classification problem. As highlighted in our hallucination analysis, this approach creates a fundamental capability gap. These models map visual features to a pre-defined and fixed vocabulary of answers (e.g., "Yes/No"). While computationally efficient, this restricts the AI from addressing long-tailed medical queries or describing unique pathologies that fall outside the training set labels. When faced with open-ended questions like "What organ contains lesions?", these models often default to their most frequent class (e.g., "Yes"), resulting in nonsensical responses.
2. **The Visual Information Bottleneck:** Standard deep learning pipelines typically resize input images to a fixed low resolution (e.g., 224x224 pixels) to satisfy the architectural constraints of pre-trained models like ResNet or ViT. In radiology, a fracture line or a nodule occupying only a few pixels cannot be missed. Our investigation into the resolution trap suggests that this

aggressive downsampling acts as a lossy compression algorithm, destroying high-frequency details. Consequently, even sophisticated language models effectively become blind to subtle pathologies.

3. **Hallucination and Grounding Failures:** Generative models, while flexible, are prone to object hallucinations (e.g., identifying a liver where there is a kidney) and plausible hallucinations (e.g., misdiagnosing appendicitis as diverticulitis because both are inflammatory). Without strong visual grounding and specific instruction tuning, models may generate text that is grammatically correct but medically factually wrong.

1.3 Research Objectives

To address these challenges, this project aims to navigate the technological shift from rigid classification to flexible and high-resolution generation. Our specific objectives are:

1. **To evaluate the performance shift across three paradigms:** We compare a traditional discriminative baseline (ResNet+BERT), a generative VLM (BLIP-2), and an instruction-tuned LVLM (Qwen-VL) to quantify the benefits of generative architectures in bridging the semantic gap between image features and textual reasoning.
2. **To analyze the impact of dynamic resolution:** We investigate how processing images at their native aspect ratios and higher resolutions affects diagnostic accuracy compared to fixed-resolution methods, specifically looking for evidence of the resolution trap in lower-resolution baselines.
3. **To assess the role of instruction tuning in hallucination mitigation:** We examine how system prompting and instruction fine-tuning influence the model's ability to reduce both low-level object hallucinations and high-level diagnostic reasoning errors.

2. METHODS

2.1 Dataset Description and Preparation

To ensure a rigorous evaluation of clinical capability, we utilized the VQA-RAD dataset which is a great benchmark in medical visual question answering. VQA-RAD consists of 315 radiological images (CT, MRI, and X-ray) and 3,515 Question-Answer (QA) pairs that have been manually verified by board-certified radiologists. As illustrated in **Figure 2.1**, the dataset contains high-resolution scans with variable aspect ratios, with many images exceeding 1000 pixels in height. This high variability presents a significant challenge for traditional fixed-resolution models. This manual curation ensures that our evaluation reflects true clinical reasoning rather than the exploitation of false statistical elements often found in synthetic data.

Image Name	Original Size (WxH)
synpic29263.jpg	(590, 480)
synpic28602.jpg	(1024, 1307)
synpic39141.jpg	(1024, 1314)
synpic53033.jpg	(415, 495)
synpic28210.jpg	(1024, 1307)
synpic41772.jpg	(759, 611)
synpic60543.jpg	(385, 514)
synpic45557.jpg	(1024, 1307)
synpic21044.jpg	(1024, 1053)
synpic41667.jpg	(1024, 1142)

Figure 2.1: Sample of original image resolutions from the VQA-RAD dataset. The data exhibits significant variance in aspect ratios and high resolutions (e.g., 1024x1307), highlighting the severe information loss that occurs when resizing to fixed 224x224 dimensions.

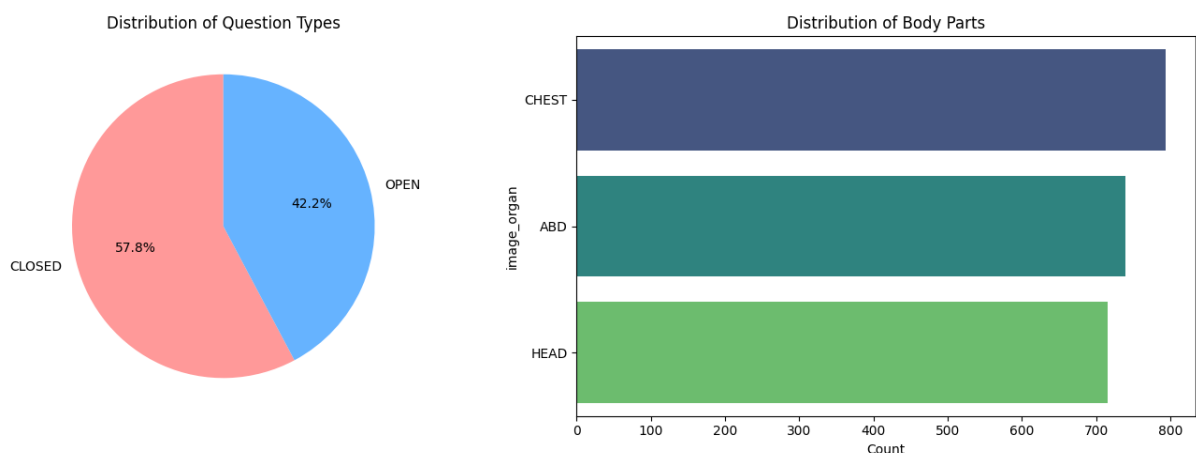


Figure 2.2: Visualization of the dataset distribution, highlighting the balance between Open-ended and Closed-ended questions, as well as the variety of body parts represented (Chest, Head, Abdomen).

To prevent data leakage where multiple slices may come from the same patient scan, we implemented a stratified split based on unique `image_id`. The dataset was partitioned into 80% training, 10% validation, and 10% testing sets. This ensures that the model is tested on unseen medical cases, assessing its generalization capability rather than its memorization.

2.2 Distinct Data Preprocessing Strategies

We employed distinct processing pipelines tailored to the specific nature of each of our chosen architectural paradigm.

A. Discriminative Baseline Preprocessing (ResNet-BERT)

Since this model treats VQA as a classification task, preprocessing focused on standardizing inputs for fixed-size tensors.

Text Processing:

- **Tokenization:** We used `BertTokenizer` (`bert-base-uncased`) with a maximum length (`MAX_LEN`) of 32.
- **Answer Mapping:** We constructed a fixed vocabulary dictionary (`ans_to_idx`) containing the 518 most frequent answers. All answers were converted to class IDs.

Image Processing:

- **Format:** All images were converted to RGB to match ResNet pre-training, even for grayscale X-rays.
- **Normalization:** Images were resized to 224x224 and normalized using ImageNet statistics (`mean=[0.485...]`, `std=[0.229...]`).
- **Data Augmentation (Training Only):** To mitigate overfitting on this small dataset, we applied:
 - `RandomRotation`: +/- 15 degrees.
 - `ColorJitter`: Random adjustments to brightness and contrast.
 - `RandomResizedCrop`: To force the model to look at different scales.

B. Generative VLM Preprocessing (BLIP-2)

For BLIP-2, we shifted to a text-generation paradigm, requiring specific prompt engineering and masking techniques.

Text Processing (Prompt Engineering):

- **Structure:** Inputs were formatted as a dialogue: "Question: {question} Answer:".
- **Closed-ended Hint:** For Yes/No questions, we appended the hint "Answer yes or no" to guide the generation.
- **Label Masking:** To train the model for Causal Language Modeling (CLM), we set the label tokens corresponding to the "Question" part to -100. This tells the PyTorch loss function to ignore the question tokens and calculate loss only on the generated answer, preventing the model from merely memorizing the input.

Image Processing:

- Images were resized to a fixed 224x224 using Blip2Processor.
- Unlike the baseline, we did not apply heavy geometric augmentations. As noted in our approach analysis, applying heavy distortion to a frozen image encoder causes a feature misalignment with the pre-trained Q-Former, degrading the model's ability to ground text in visual content.

C. Large Multimodal Preprocessing (Qwen-VL)

To address the resolution trap, we implemented advanced dynamic processing.

Dynamic Resolution & Vision Info:

- Instead of a fixed resize, we utilized `qwen_vl_utils.process_vision_info`.
- We set `min_pixels = 256x28x28` and `max_pixels = 384x28x28`. The model dynamically cuts images into variable numbers of patches based on their native aspect ratio. This preserves fine-grained details in long spinal X-rays that would be lost in a square resize.

Chat Template:

- We utilized the specific chat template required by Qwen-Instruct:

- System Role: "You are an expert radiologist..." (Setting the expert persona).
 - User Role: <image> {question}.
 - Assistant Role: {answer}.
- Similar to BLIP-2, we applied masking (-100) to the system and user prompts, ensuring the model is trained only on the assistant's reasoning.

2.3 Model Architectures and Implementation

We designed our experiments to compare three distinct stages of VQA evolution.

A. The Discriminative Baseline (ResNet-50 + BERT)

To represent the traditional classification approach, we constructed a joint embedding architecture that processes visual and textual inputs in parallel streams. The visual stream employs a ResNet-50 backbone pre-trained on ImageNet. We modified this network by removing the final fully connected layer to extract high-level spatial features, which are then projected into a 512-dimensional space via a linear layer. Simultaneously, the textual stream utilizes a pre-trained BERT (bert-base-uncased) to encode the questions. We specifically extracted the [CLS] token embedding, which represents the semantic summary of the input sentence, and projected it to a matching 512 dimensions. These visual and textual features are fused via concatenation to form a 1024-dimensional vector, which is then passed through a Multi-Layer Perceptron (MLP) classifier. This classifier consists of a linear layer, a ReLU activation, and a Dropout layer (0.3) to prevent overfitting, ending with a final output layer mapped to 518 classes. However, this architecture is mathematically restricted by this fixed output vocabulary, rendering it incapable of generating novel answers for open-ended queries outside its training set.

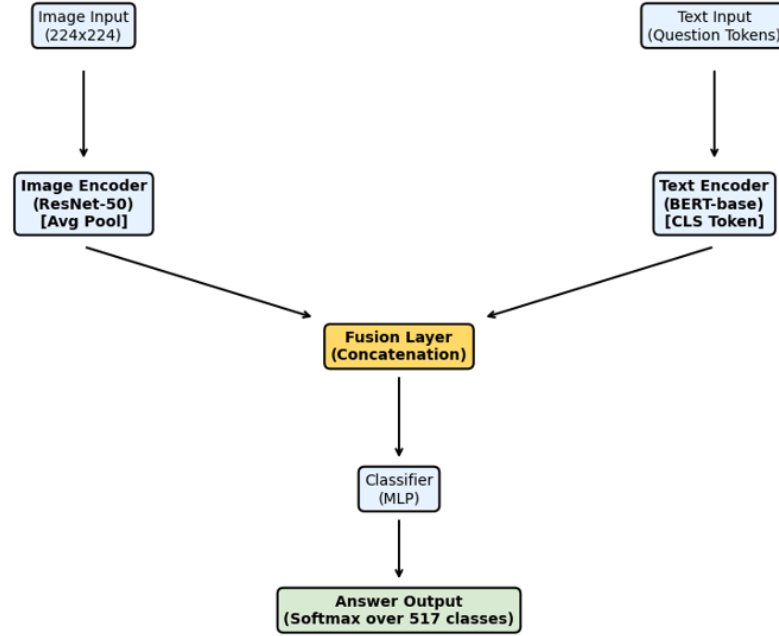


Figure 2.3: Schematic of the Baseline CNN-BERT architecture. The model fuses ResNet-50 visual features and BERT textual embeddings via concatenation to predict answers from a fixed vocabulary.

B. The Generative VLM (BLIP-2)

Representing the shift to generative AI, we deployed BLIP-2 (Bootstrapping Language-Image Pre-training). This architecture consists of a frozen image encoder and a frozen large language model (OPT-2.7b), bridged by a lightweight querying transformer (Q-Former). The Q-Former acts as an information bottleneck which extracts the most relevant visual features conditioned on the text query.

Fine-tuning a 2.7 billion parameter model is very computationally expensive. To mitigate this, we applied Low-Rank Adaptation (LoRA) to the attention layers (q_proj , v_proj , k_proj) of the language model. By setting the rank $r = 16$ and alpha $\alpha = 32$, we made only 0.56% of the parameters trainable, allowing for efficient adaptation to the medical domain without serious forgetting issues.

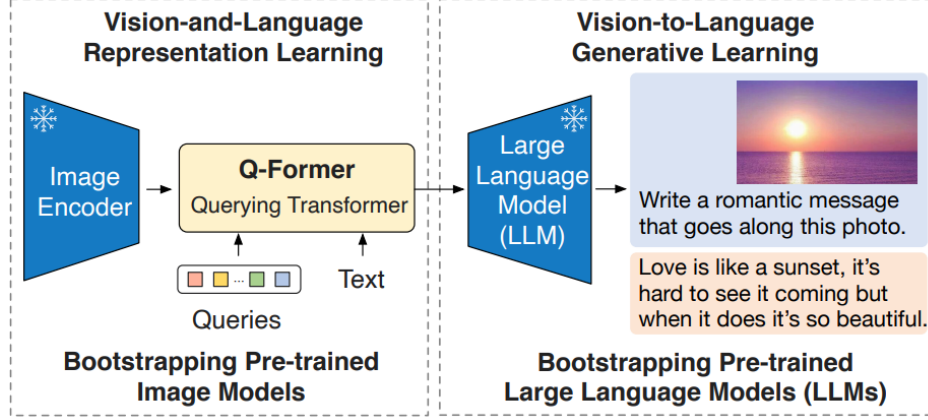


Figure 2.4: Overview of the BLIP-2 framework. The architecture utilizes a Querying Transformer (Q-Former) to bridge frozen image encoders and large language models.

C. The Instruction-Tuned LVLM (Qwen-VL)

To address the resolution constraints and hallucination issues of previous models, we utilized Qwen-VL-Chat (Int4 version). Unlike BLIP-2, which relies on a complex Q-Former, Qwen-VL employs a simpler direct adapter to project visual tokens into a Decoder-only Transformer architecture. The defining feature of this implementation is its support for flash attention and long-context windows, which are essential for handling the dynamic resolution input strategy described in Section 2.2. By supporting variable sequence lengths, the model can process images split into a flexible number of patches, allowing it to see fine-grained details in high-resolution medical scans without the information loss caused by fixed-size resizing. Furthermore, the model's instruction-tuning capabilities allow it to strictly follow system prompts, adopting the persona of an expert radiologist to improve response professionalism.

2.4 Training Strategy

Training was conducted on a single NVIDIA T4 GPU (16GB VRAM). We deployed specific memory optimization strategies:

1. Baseline Two-Stage Training:

- Stage 1 (Epoch 1-3): We set `requires_grad=False` for ResNet and BERT, training only the classifier head. This prevents random initialization noise from the classifier disrupting the pre-trained feature extractors.
- Stage 2 (Epoch 4+): We unfroze all layers (`requires_grad=True`) but reduced the learning rate to $1e-5$.
- Scheduler: ReduceLROnPlateau (`factor=0.1`, `patience=2`) monitoring validation loss.

2. Generative Models:

- **Quantization:** We employed distinct strategies based on model size. For Qwen-VL, we utilized QLoRA, loading the model in 4-bit precision (nf4) via `BitsAndBytesConfig` to accommodate its larger parameters and dynamic resolution. For BLIP-2, we applied standard LoRA without quantization, loading the base model in half-precision (float16).
- **Gradient Accumulation:** Due to the high resolution of Qwen-VL, we were limited to a batch size of 2. We implemented `gradient_accumulation_steps=8` to achieve an effective batch size of 16, stabilizing updates.
- **Optimizer:** We used `PagedAdamW8bit`, which offloads optimizer states to CPU RAM to prevent Out-Of-Memory (OOM) errors.

2.5 Evaluation Metrics

We assessed performance using both classification and generation metrics:

1. **Closed-ended Accuracy:** Accuracy on Yes/No or limited-choice questions. For generative models, we implemented a post-processing cleaning function (e.g., converting "The answer is yes" to "yes") to ensure fair comparison with ground truth labels.
2. **Open-ended Accuracy:** Exact string-matching accuracy. We employed this metric because the open-ended answers in the VQA-RAD dataset are typically short and concise (often 1-3 words).

3. **BLEU & ROUGE-L:** To account for semantic correctness where exact string-matching fails (e.g., "Left lung" vs. "Lung (left)"), we calculated BLEU (n-gram precision) and ROUGE-L (longest common subsequence).
4. **Confusion Matrix:** We analyzed the false negative rates to determine clinical risk.

3. RESULTS

3.1 Quantitative Performance Analysis

We conducted a comprehensive evaluation of the three architectural paradigms on the VQA-RAD test set. Table 1 summarizes the performance across closed-ended (Yes/No, specific choice) and open-ended (generative) questions.

Table 3.1: Overall Performance Comparison

Metric	Baseline (ResNet+BERT)	BLIP-2 (LoRA)	Qwen-VL (QLoRA)
Paradigm	Discriminative	Generative	Instruct-LVLM
Input Resolution	Fixed (224×224)	Fixed (224×224)	Dynamic (Max 384px)
Closed-ended Acc	59.48%	57.76%	77.59%
Closed-ended F1	64.86%	58.41%	78.95%
Open-ended Acc	5.32%	22.34%	32.98%
Open-ended ROUGE-L	5.74	32.09	43.79%
Inference Speed	High (Milliseconds)	1.58 it/s	1.13 it/s

The results reveal a critical and counter-intuitive phenomenon we term the resolution trap. The generative BLIP-2 model, despite possessing a sophisticated Q-Former and LLM, underperformed the simpler discriminative baseline in closed-ended tasks (57.76% vs. 59.48%). This regression indicates that when input resolution is blindly restricted to 224x224, the generative model loses the visual evidence required for binary diagnostic decisions. Conversely, Qwen-VL achieved a dominant 77.59% accuracy (an ~18% leap). This improvement is directly attributable to its dynamic resolution mechanism, which processes images at their native aspect ratios and preserving the fine-grained details of X-ray fractures and nodules that are otherwise destroyed by aggressive resizing.

In open-ended tasks, the baseline hit a paradigm wall, achieving only 5.32% accuracy due to its fixed vocabulary constraints. It simply cannot answer questions outside its pre-defined 518 classes. Qwen-VL not only achieved the highest exact-match accuracy (32.98%) but also a superior ROUGE-L score of 43.79. This metric highlights that even when Qwen-VL did not match the exact ground truth string,

its generated responses were semantically aligned and medically relevant, demonstrating robust knowledge.

This performance comes at a cost. Qwen-VL operates at 1.13 iterations/second, making it approximately 28% slower than BLIP-2 (1.58 it/s) and orders of magnitude slower than the baseline. While acceptable for a diagnostic aid, this latency presents a challenge for real-time emergency triage applications, necessitating a trade-off between extreme accuracy and instant response.

3.2 Training Dynamics and Convergence

To understand the stability and confidence of these models, we analyzed the training loss curves.

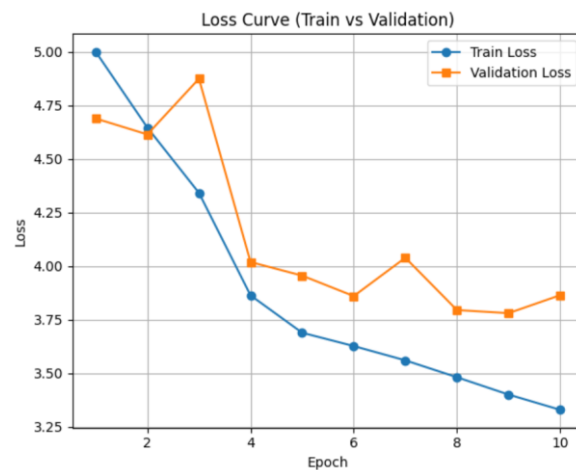


Figure 3.1: Loss curve of the baseline model.

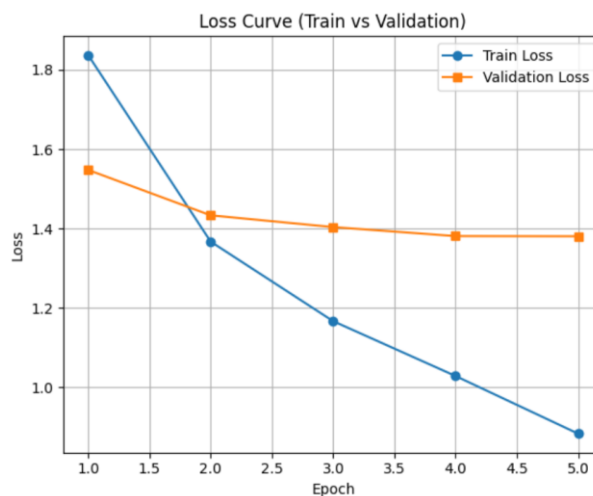


Figure 3.2: Loss curve of the BLIP-2 model.

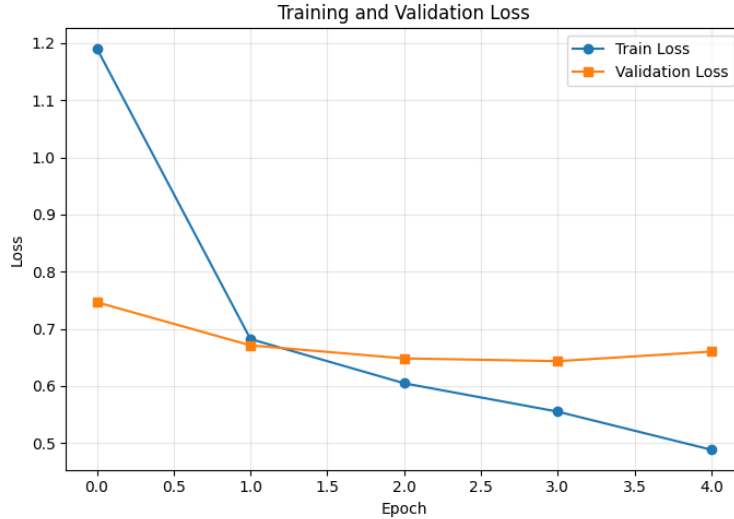


Figure 3.3: Loss curve of the Qwen-VL model. The smooth decrease in loss demonstrates the stability provided by PEFT (LoRA).

All models exhibited signs of overfitting due to the small size of the VQA-RAD dataset. However, the nature of the training dynamics differed significantly. The baseline which used full fine-tuning exhibited a jittery loss curve, indicating high sensitivity to batch noise. In contrast, BLIP-2 and Qwen-VL, trained via PEFT (LoRA), showed much smoother convergence. This validates that optimizing only a small subset of parameters (0.56%) acts as an effective regularizer, preventing the catastrophic forgetting often seen in full fine-tuning.

We also observed a massive disparity in the minimum validation loss:

- **Baseline:** Plateaued at around 3.78
- **BLIP-2:** Descended to around 1.38
- **Qwen-VL:** Descended to around 0.64

This order of magnitude difference implies a profound confidence gap. The baseline remains high in perplexity, hesitant in its predictions, whereas Qwen-VL establishes a high degree of certainty in its reasoning.

3.3 Confusion Matrix Analysis

We analyzed the confusion matrices for closed-ended questions to evaluate clinical safety, specifically focusing on false negatives (Type II errors) which represent missed diagnoses.

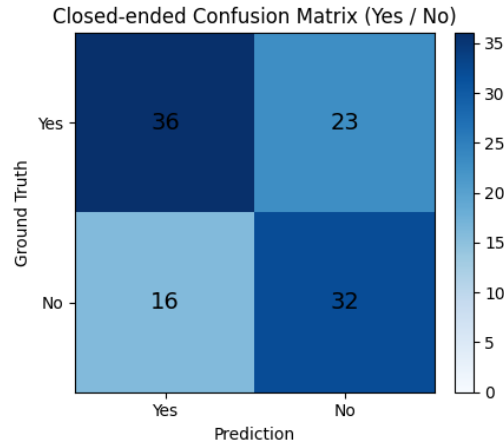


Figure 3.4: Confusion Matrix for baseline ResNet-BERT. High rate of false negatives (23 cases).

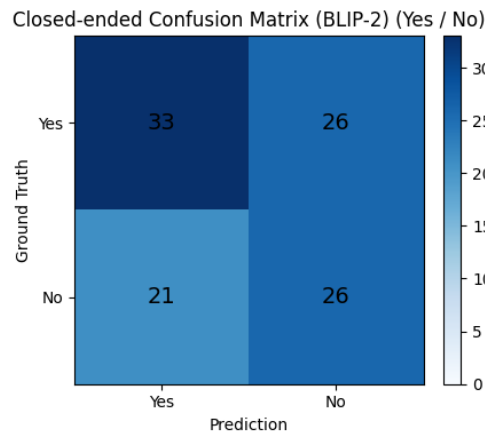


Figure 3.5: Confusion Matrix for BLIP-2. High rate of false negatives (26 cases).

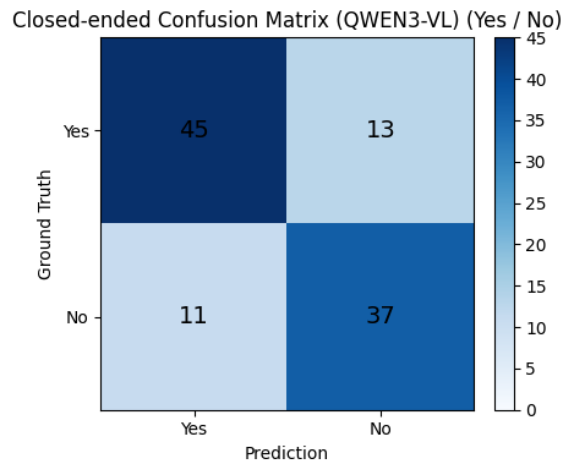


Figure 3.6: Confusion Matrix for Qwen-VL. Significant reduction in False Negatives (13 cases).

Qwen-VL demonstrated a superior safety profile by significantly reducing the rate of missed diagnoses. BLIP-2 misclassified 26 positive cases as negative, performing worse than the baseline which misclassified 23 cases. This regression further confirms the resolution trap, where low-resolution inputs

effectively blind the model to small lesions. Qwen-VL which leveraged dynamic resolution, reduced false negatives to 13 cases. This is a 50% reduction in risk compared to BLIP-2. The strong diagonal dominance in Qwen-VL's matrix indicates a high discriminative power, proving that high-resolution inputs allow the model to correctly identify positive findings that other models miss.

Visual analysis of the matrices shows that Qwen-VL exhibits strong diagonal dominance, marked deep blue on the TP/TN axis, whereas the baseline and BLIP-2 show lighter colors and more spread. Qwen-VL demonstrated high sensitivity by identifying lesions and high specificity by correctly excluding healthy cases, largely eliminating the over-generation issue where generative models hallucinate problems in healthy scans.

3.4 Qualitative Case Studies

To move beyond aggregate metrics, we analyzed specific failure modes to understand the cognitive behaviours of the models.

A. The Semantic Gap (Baseline vs. Generative)

We examined a case where the model was asked to identify a brain structure ("In what ventricle is the enhancing mass seen?").

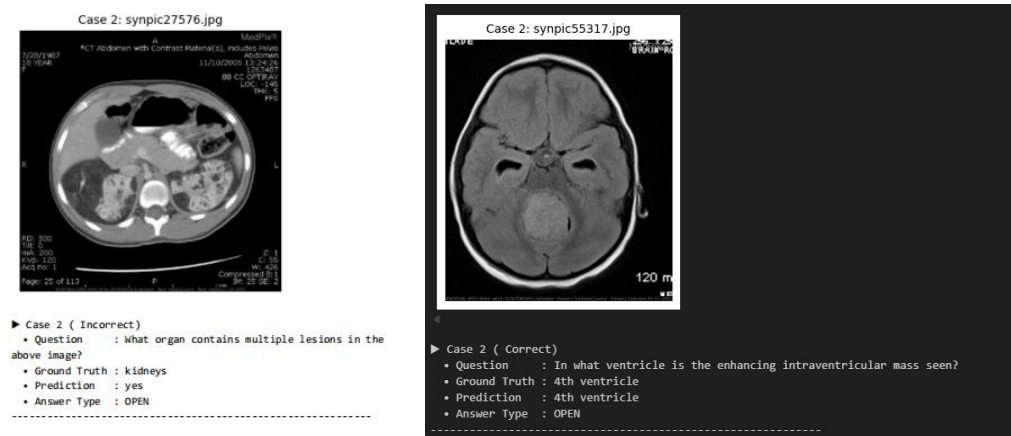


Figure 3.7: Comparison of baseline vs. Qwen-VL answer. The baseline model fails to understand the open-ended question, defaulting to a classification label "Yes", whereas Qwen-VL provides a semantically accurate anatomical location.

- **Observation:** The discriminative baseline predicted "Yes."

- **Analysis:** This error is mainly because the baseline has hit its paradigm wall. Because the baseline treats VQA as a classification task, when faced with a complex query outside its label distribution, it biases towards the most frequent class ("Yes/No"). It fundamentally lacks the language modelling capability to form a semantic answer. In contrast, Qwen-VL correctly generated specific anatomical terminology ("4th ventricle"), proving that generative architectures are the only viable path for open-ended medical inquiry.

B. Visual Grounding and Object Hallucination

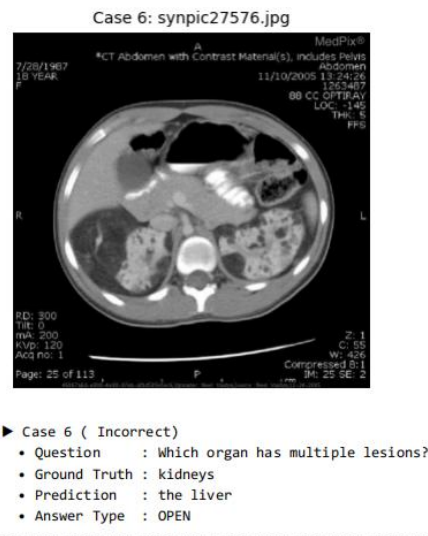


Figure 3.8: Visual Grounding Failure. BLIP-2 detects the pathology (cysts) but hallucinates the location as "Liver" instead of "Kidneys," highlighting a lack of spatial grounding.

- **Case Study:** In an abdominal CT scan showing renal cysts.
- **Observation:** BLIP-2 correctly identified "Lesions" but located them in the "Liver" instead of the "Kidneys."
- **Analysis:** BLIP-2 correctly understood the type of question where it named an organ but failed the visual grounding. It exhibited object hallucination, misidentifying the kidneys as the liver. This suggests BLIP-2 relied on language priors (e.g., liver cysts are common) rather than visual evidence, likely due to feature misalignment from the fixed resolution. Qwen-VL, leveraging dynamic resolution and instruction tuning, correctly grounded the visual features to the correct anatomical entity.

C. Fine-Grained Spatial Reasoning

We observed clinical feature confusion in complex differential diagnoses.

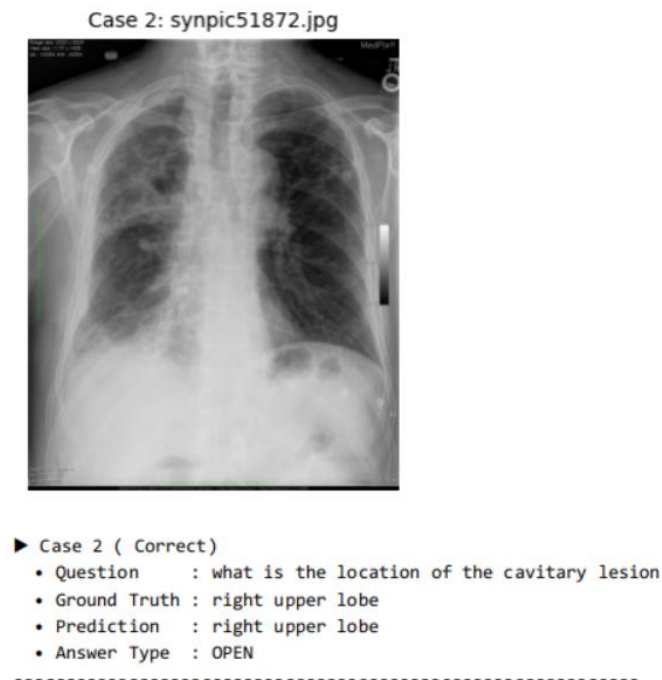


Figure 3.9: Cavitory Lesion Localization of Qwen-VL model.

- **Observation:** The model not only identified the cavitory lesion but also pinpointed its precise anatomical location. Qwen-VL generated the exact ground truth: "right upper lobe."
- **Analysis:** Spatial localization is a common failure point for VQA models, which often confuse left and right or hallucinate lobes due to lack of visual clarity. This success demonstrates Qwen-VL's superior visual alignment. By processing the image at high resolution without distortion, the model successfully correlated the visual texture of the lesion with the correct anatomical coordinates, proving it has overcome the spatial disorientation issues observed in lower-resolution baselines.

4. DISCUSSION

4.1 Discriminative Efficiency and Generation Capabilities

Our results highlight a distinct trade-off between task alignment and semantic capability, confirming that newer models are not always better for every sub-task.

A. The Efficiency of the Discriminative Baseline

A critical anomaly in our data was the discriminative baseline (59.48%) outperforming the generative BLIP-2 (57.76%) on closed-ended questions. This phenomenon which we term discriminative efficiency, can be attributed to task alignment.

- **Inductive Bias:** For binary tasks (Yes/No), discriminative models possess a natural inductive bias. They simply learn a decision boundary that maps high-level feature vectors directly to a class index.
- **Data Distribution Exploitation:** The VQA-RAD dataset is heavily skewed towards "Yes/No" answers. The baseline, operating as a classifier, efficiently minimizes cross-entropy loss to fit this long-tailed distribution's high-frequency head. It does not need to understand language, it only needs to correlate visual patterns with a specific label ID.

B. The Generation Overhead of GenAI

In contrast, BLIP-2 bears the burden of generation overhead. Even for a simple "Yes/No" question, a generative model must calculate the probability distribution of the next token across a massive vocabulary. When combined with the resolution trap (fixed 224x224 input blurring visual features), this complexity becomes a liability. The generative model attempts to construct a sentence based on blurry visual evidence, whereas the classifier takes a shortcut to the most probable label.

C. The Semantic Crossing

However, the baseline hits a hard paradigm wall on open-ended tasks where it scores only 5.32% accuracy on unseen answers. This confirms that while discriminative models are efficient specialists for

fixed-scope diagnostics, Generative VLMs (like Qwen-VL) are necessary generalists capable of flexible reasoning, provided they can overcome the resolution barrier.

4.2 The Visual Information Bottleneck

The most critical engineering finding of this study is the decisive role of dynamic resolution. The resolution trap where the generative BLIP-2 model (57.76%) underperformed the simpler Baseline (59.48%) reveals that treating medical images like natural images (resizing to 224x224) is a fatal flaw in radiology. Medical pathologies are often defined by high-frequency details, such as a fracture line or a texture change, that occupy only a fraction of the image pixels. Standard resizing acts as a lossy compression algorithm that erases this evidence. Qwen-VL's ability to process images at native aspect ratios allows it to see these details, directly correlating with its 50% reduction in false negatives. This suggests that in medical AI, input quality is often more critical than model size.

4.3 Hallucination and Clinical Reliability

Our qualitative analysis reveals that model errors are not uniform. Rather, they evolve in complexity as model architecture advances. We categorize this evolution into three distinct stages, ranging from fundamental semantic failures to high-level reasoning errors.

A. The Capability Gap

The first hurdle in Med-VQA is the capability gap between discriminative and generative paradigms.

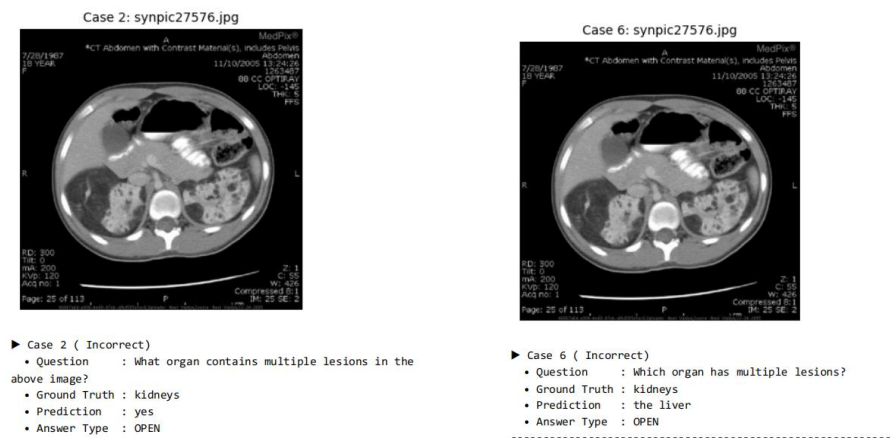


Figure 4.1: Comparison of answers to open-ended questions of baseline (left) and BLIP-2 (right).

- **The Semantic Void (ResNet-BERT):** As shown in **Figure 4.1**, when asked "What organ contains multiple lesions?", the baseline predicted "yes". This confirms that discriminative models do not hallucinate in the traditional sense. Rather, they hit a paradigm wall where they cannot process open-ended semantics, defaulting to the most frequent training label.
- **Object Hallucination (BLIP-2):** BLIP-2 successfully overcame this barrier, correctly identifying that the question required an anatomical noun. However, it predicted "the liver" when the ground truth was "kidneys." This represents a classic object hallucination. The model likely relied on language priors (e.g., the statistical frequency of liver lesions in the dataset) rather than visual evidence, creating a grammatically correct but factually wrong answer.

B. Visual Alignment and Grounding

The second stage of evolution is the improvement in visual grounding, where the model aligns textual concepts with specific visual features.

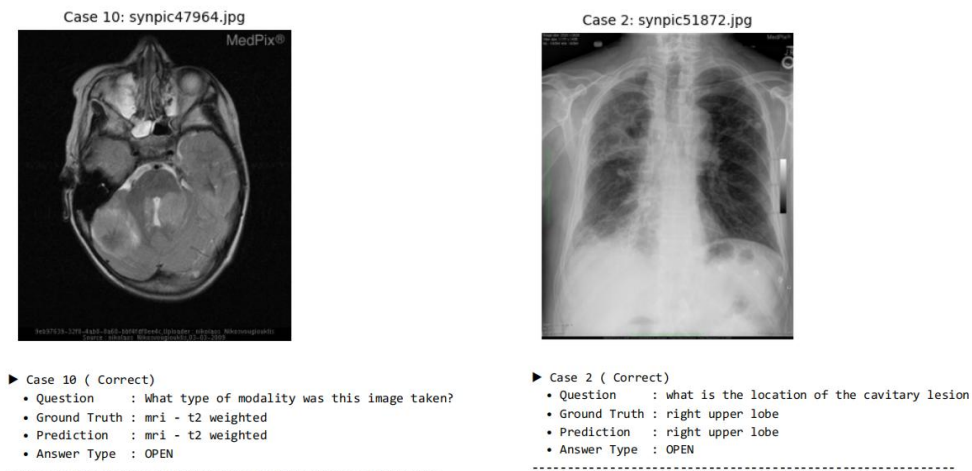


Figure 4.2: Answers to open-ended questions by Qwen-VL.

- **Reduction in Blind Guessing:** In contrast to BLIP-2's tendency to hallucinate common organs, Qwen-VL demonstrated superior alignment. In **Figure 4.2**, Qwen-VL accurately identified the image as "mri - t2 weighted".
- **Significance:** This specific prediction identifying a T2-weighted sequence cannot be guessed via language probability alone. It requires the model to physically see and interpret fluid signal

intensities. This indicates that Qwen-VL has significantly reduced low-level hallucinations by effectively grounding its language generation in visual data.

C. Plausible Hallucination

While Qwen-VL eliminates simple object errors, it introduces a more complex challenge: plausible hallucination.

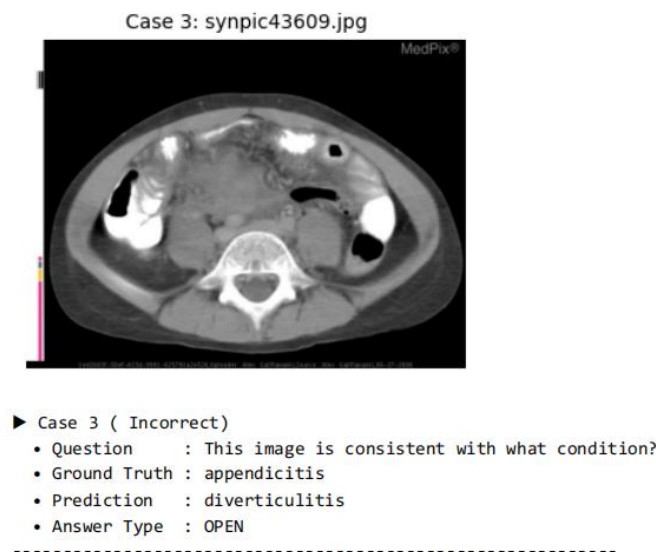


Figure 4.3: Ground truth was ‘appendicitis’, but Qwen-VL predicted ‘diverticulitis’.

- **Case Study:** In **Figure 4.3**, the model was presented with a CT scan showing lower abdominal inflammation. The ground truth was "Appendicitis," but Qwen-VL predicted "Diverticulitis".
- **Why this is "Plausible":** Unlike the baseline's random guess ("yes") or BLIP-2's unrelated object error, this error is medically logical. Both appendicitis and diverticulitis present as inflammatory processes in the bowel with similar visual features (fat stranding, wall thickening).
- **Conclusion:** This indicates that Qwen-VL has achieved expert perception by detecting the inflammation but struggles with expert reasoning where it needs to differentiate fine-grained pathologies based on specific anatomical location. This plausible hallucination represents the current boundary of large multimodal models in medicine. The model is no longer blind, but it is not yet fully clinical.

4.4 Limitations and Trade-offs

The superiority of Qwen-VL comes with distinct trade-offs that must be considered for deployment:

- **Accuracy-Latency Trade-off:** Qwen-VL operates at ~ 1.13 it/s, significantly slower than the millisecond-level inference of the baseline. In a high-volume emergency room triage scenario, this latency might be a bottleneck, necessitating a balance between diagnostic depth and speed.
- **Data Scarcity:** While QLoRA prevented catastrophic forgetting, the model's struggle with complex differential diagnoses (Appendicitis vs. Diverticulitis) suggests that the small VQA-RAD dataset (3,515 pairs) is insufficient for teaching expert reasoning. The model learns what to look for, but not always how to distinguish subtly similar pathologies.
- **2D vs. 3D Context:** The spatial disorientation observed in some cases (confusing adjacent organs) highlights the limitation of using 2D image-text pre-training for 3D volumetric data (CT/MRI). Without volumetric context, the model lacks the spatial priors inherent to human radiologists.

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

This study has systematically navigated the architectural evolution of Medical Visual Question Answering, advancing from rigid discriminative baselines to state-of-the-art large Vision-Language Models. Our comprehensive evaluation on the VQA-RAD dataset yields three definitive conclusions regarding the deployment of AI in radiology.

First, we have confirmed the paradigm wall facing traditional deep learning. The discriminative baseline’s inability to generate novel answers (5.32% accuracy on unseen open-ended queries) renders it unsuitable for the complex and variable nature of clinical dialogue, despite its computational speed. Second, we identified a critical resolution trap in standard generative models. The BLIP-2 architecture, while theoretically superior, failed to outperform the baseline on closed-ended diagnostic tasks (57.76% vs. 59.48%). This failure is directly attributable to the fixed 224x224 input resolution, which acts as an information bottleneck, filtering out the high-frequency visual evidence necessary for detecting subtle pathologies.

Third, Qwen-VL represents a viable path forward for the AI radiologists. By breaking the resolution bottleneck with dynamic encoding and utilizing instruction tuning, it achieved a state-of-the-art accuracy of 77.59% and most importantly, improved clinical safety by reducing the false negative rate by 50%. However, we discovered that while Qwen-VL effectively mitigates low-level object hallucinations, it remains susceptible to plausible hallucinations which are medically coherent but incorrect diagnoses (e.g., confusing Appendicitis with Diverticulitis). This indicates that while the model has mastered medical perception, it still requires further refinement to master complex differential diagnostic reasoning.

5.2 Future Work

While Qwen-VL demonstrates strong reasoning capabilities, our error analysis points to specific areas for future optimization:

- **Retrieval-Augmented Generation (RAG):** To address plausible hallucinations in differential diagnosis, future systems should integrate external medical knowledge bases (e.g., Radiopaedia). RAG would allow the model to cross-reference visual features with clinical guidelines before generating a diagnosis, bridging the reasoning gap.
- **3D Volumetric Encoders:** Our study revealed that 2D models struggle with spatial disorientation in axial slices (e.g., confusing liver and kidney). Integrating 3D-CNN backbones or video-based transformers could provide the volumetric context necessary to distinguish adjacent organs in CT/MRI scans.
- **Knowledge Distillation:** To address the accuracy-latency trade-off, we propose distilling the reasoning capabilities of the 4-billion parameter Qwen-VL into a smaller student model. This would aim to retain the semantic understanding of the LVLM while recovering the millisecond-level inference speed of the baseline, enabling real-time deployment in emergency settings.

6. REFERENCES

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7. APPENDIX

Project Repository:

The complete source code for data preprocessing, model training (Baseline, BLIP-2, Qwen-VL), and evaluation scripts is available at the following GitHub repository:

<https://github.com/JianHengHin0831/Advanced-Machine-Learning-Alternative-Assessment>

Note: The repository includes the Jupyter Notebooks used to generate the loss curves and confusion matrices presented in this report.

8. MEMBER CONTRIBUTIONS

Name	Contribution	Weightage
Hin Jian Heng	Responsible for the end-to-end implementation and training of all three models (ResNet-BERT Baseline, BLIP-2, and Qwen-VL). Conducted the comprehensive performance evaluation, generated all experimental data (loss curves, confusion matrices), and derived the key technical findings regarding the resolution trap and hallucination patterns.	50%
Tee Chee Hong	Responsible for the complete documentation lifecycle, including writing and structuring both the preliminary project report and this final report. Synthesized technical findings into the academic narrative and prepared the presentation slides and materials for the project.	50%